

Section 1: The Structure of a Technical Report



Figure 1. Technical reports typically follow similar structure and format.

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In this section we will examine the specific goals and purposes of each section of a technical report. For the purposes of PD 11, you need to understand the rules and structure for writing a technical report. Keep in mind that once you become a full-time working professional, the rules for writing will likely change as your audience changes.

Sections of a Technical Report

The major sections of a technical report include a title page, letter of submittal, executive summary, table of contents, list of figures and tables, introduction, body (i.e., literature review/background, materials and methods, results and analysis, discussion), conclusion, recommendations, references, and appendix.

As we are going to discuss below all the sections of a technical report, we encourage you to revisit the award-winning example technical reports written by your fellow students we mentioned in the previous unit, and see how they followed the structure of a technical report:

- **Sijia Wang (Statistics) - The Importance of Standardized Interfacing for Manipulating Scientific Data (PDF)**
- **Amy Chan (Biochemistry) - Behavioural and Psychological Symptoms of Dementia in Memory Clinic (PDF)**
- **Sarah Odinotski (Nanotechnology Engineering) - Optimization of Cell Plating Density for the Early Establishment of Pancreatic Ductal Adenocarcinoma Organoids (PDF)**

Title Page

Should include, in order: University of Waterloo, your Faculty, the title of your report, the name and location of your employer, your name and ID number, your program, the date when the report was prepared. We will discuss this in more detail in the next section.

The title page conveys essential information about the report including who wrote it and what it is about. The title page should show the:

- Name of the University and the Faculty you belong to
- Title of the report
- Name and location of the workplace
- Key information about who the report was prepared by: student name, student

number, year and major, and the date of submission

Letter of Submittal

The letter of submittal should follow immediately after the title page and should include the title or subject of the report, the reason it was written, as well as other information that we detail below.

The letter of submittal is used to preface a longer text and is addressed directly to the reader of that longer text. It should include the:

- Report title
- Employer's location and main activity of the department (if applicable)
- Purpose of report
- Acknowledgments
- Statement of endorsement
- Your name, student ID number, and signature

Executive Summary

The executive summary summarizes key information contained within the report. This section should include key points about the purpose of the report as well as key findings.

Table of Contents

The table of contents reveals to the reader what the organization of the report will be. When creating digital documents, the table of contents can also contain dynamic links that allow the reader to jump from the table to specific sections of the report.

List of Figures and Tables

The list of figures and tables is used in technical reports to share data in an accessible way (i.e., page numbers provided where they can be located in the report). Figures and tables must always have a caption and/or legend that is sufficiently detailed and referenced so the reader understands the data presented and where it came from.

Introduction

An introduction should describe the project or problem, and clearly state the overall objective(s) of the report. It is a report convention that makes predictions about patterns and possibilities associated with the subject matter or content of the report. An introduction is different than a summary or abstract because it is the beginning of the body.

Body

You wouldn't necessarily label this portion of the report the "body". Instead, the body usually consists of headings and/or sub-headings to take the reader through key processes or key steps. Standard headings used in the body include:

- Literature Review
- Materials and Methods
- Results and Analysis
- Discussion

The headings could be numbered and might include sub-headings to break up the body into parts and to describe each section. The body of the report is where you apply your analysis to a workplace issue or problem. The goal is to perform an analysis that reveals new patterns, conclusions, oversights, or opportunities in the workplace. This analysis needs to be described in a detailed and technical way so that it can be

reproduced and repeated in the future. Again, this portion of the report is where you write up the technical work you have done in a way that allows others to reproduce it.

Conclusion and Recommendation

The conclusion summarizes the key takeaways and messages contained within the technical report. The conclusion statements need to build on the analysis, highlighting the key points or answering the main questions posed by the report. Conclusion statements need to be specific and concise. This is not a place to introduce new ideas or analyses. Avoid using generalities in this section.

The recommendations section provides the reader with actionable items to implement based on the findings of the report and is often presented immediately following the conclusion. These recommendations should tell the reader exactly what to do. A recommendation is like a recipe and needs to include all relevant details about the who, what, when, where, why, and how of what should happen next, so that there is no ambiguity about any of the directions, dates, or deadlines.

References

The references section is a full listing of all sources cited in the report. For the purposes of this course, you are required to use [APA Style 7th edition](#)  format for both the reference list and in-text citations.

Appendix

An appendix is a space for attaching data, specific references, and supplementary materials that readers need to have access to in order to understand your report, to reproduce your findings, or to check your analysis. When these materials are lengthy or simply would take a long time for a reader to review, they should not be included in

the body of the report. The appendix is a section at the back of the report where you can include lengthy materials in full. You show consideration for your audience by making them available - and by not interrupting the flow of the report by including them in the body.

Recognizing these key sections of a technical report allows you to recognize key features of the work you'll do in co-op. For example, you may be expected to bring analytical skills to bear on workplace issues or you may be expected to develop and advance technical expertise in the workplace. In a technical report, this expertise should be conveyed through figures and tables that make data available.

Based on the target audience, technical reports may require different formatting depending on your employer standards/guidelines. It is similar to submitting an article to a journal: each journal has its own formatting requirements and style (e.g., headings, sections, font style, paragraph, margins, page numbering style).

For example, for any audience, the use of headings is critical for organizing and breaking up content in a technical report. But different audiences (and employers) may expect headings to be written in different ways.

Waterloo Writing Works: Headings

A heading is usually a single word or a short phrase that tells the reader what the following section will be about. In doing this, you will make your technical report much easier to read.

For your interest, select the button to learn more about how to write and use headings for your technical report.

[Headings](#) 

For your Technical Report, you will use the **PD 11 Technical Report Guidelines (PDF)** . These guidelines include full details on what should be included in each section for your technical report.

Review PD 11 Technical Report Guidelines to identify the type of content required for these sections. Reviewing the example reports also really helps.

Also, to help you jumpstart your writing, we are providing the **First Draft of Technical Report Template (DOCX)** document. You may use it as you start drafting your report this week. **Note:** clicking the link above will initiate the download of the DOCX file onto your computer.

Faculty Discuss Structure of Technical Reports

We asked several University of Waterloo faculty members to comment on some of the ways that the structure of a technical report impacts their reading of reports. We also asked them to offer their best advice about formatting reports.

Watch the video for advice from faculty members on structuring a technical report.

Faculty Advice on Technical Report Structure

[Video can be viewed in LEARN.]

Video 1. Structure Benefits the Reader. In this video, several Waterloo professors comment on how the structure of a technical report benefits their reading and understanding of the report contents. Credit: © Course Author(s) and University of Waterloo.

Transcript



[Text only; no audio] Waterloo professors will talk about formatting conventions

and guidelines for reports.

[Text only; no audio] To begin with, professors like Dana Church from the School of Pharmacy can tell you that templates are as much a part of the work world as they are a part of the report process:

Dana Church]: So my name's Dana. I'm a research coordinator with University of Waterloo School of Pharmacy, and I work specifically with an organization called OPEN, the Ontario Pharmacy Research Collaboration. And I work on their Pharmacists as Immunizers project.

We actually have quite a few templates in my work. They can range anywhere from templates for PowerPoint presentations, for conferences, all the way to even agenda forms. So when you're making agenda for a meeting, we have a specific template. Also for reports itself, so we have a template that has the title page, table of contents, and all the different chapters or sections.

And they all look the same. For our purposes, they all have our organization's logo, they all have the colours. And at first I thought it was rather strange that we had to use all these templates that look the same, but actually I find them really handy because that way, just by glancing at the pile of papers on my desk, even I can tell what these documents are, because of what they look like. And also, if you have more than one person writing reports, it just gives the reports consistency.

So, all of the reports are laid out similarly, you're not having to figure out how each report is organized, per se, for each one. So it's nice for consistency, and recognizability, for sure. And it makes work more efficient in a lot of ways.

So, I've used templates in other positions as well. Definitely, where I'm working now, they have more, not so much complex, but they have more detailed templates and different types of templates. But, pretty much the similarities that I see between them are that they use the, let's say, if an organization has a certain colour or a certain logo, they're always on them.

Similarities, in terms of organizing the report, so basically they all start with a title

page, or just a generic title page, table of contents, and then how the chapter headings or section headings and subheadings, the font, the size of the font, how the pages are numbered, those tend to be similar. And, you know, it might sound very minor and superficial to have such consistency, but it actually does really help to have consistency across those reports.

Then I've noticed, working in different organizations that, that yeah, they generally tend to have similar organization to the different reports. Again, it helps so that you're not reinventing the wheel every time you go to sit down to write a report. It's basically filling in a template, rather than trying to design a whole report yourself.

[Text only; no audio] Remember: The structure of the report impacts the actual process of opening the report and reading it.

[Steven Furino]: So, my name is Steven Furino. I am the Associate Dean for Undergraduate Studies in the Faculty of Mathematics. So when I receive a report, the first thing I expect to see is a basic structure of an executive summary, the body of the report, and the set of appendices that contain the data or the fine details that were required to make the assessments that appear in the body and the executive summary.

The first thing I do is read the first paragraph of the executive summary. I scan the rest of the executive summary, hoping, expecting, that that will direct me into the parts of the body that are most important for the decision at hand. That may depend on the value, the controversy, unresolved issues. Then I'll go to the appendices if needed, and I assume that the author of the report will have directed me to the appropriate places in the appendices.

[Text only; no audio] The structure of the report also reflects values of the discipline: science reports represent scientific methods, and likewise for math reports.

[Tadeusz Gorecki]: So my name is Tadeusz Gorecki. I'm the Associate Dean for co-op in the Faculty of Science. In science, in a way, marking is a relatively simple process because we actually have a fixed format that we require

students to use and there's a reason for it. There is a reason why all papers that are published in scientific literature have very similar structure because that is really helpful to the reader.

So the first thing I look at is the structure, and simply following the rubric that we have prepared for marking, where we look for all the parts required. Are they there? Do they meet the expectations that we put forward? What is the overall impression of the report? And then, of course, the next step is sitting down and reading the whole thing, marking any errors that I may find or writing suggestions on the report, what can be improved.

So when it comes to the structure, like I mentioned, just like any scientific paper, the format is pretty straightforward. You start with an introduction and the role of the introduction is to literally introduce the reader to the topic, to paint a broad picture of what it is that you are dealing with and what was your actual project, what you were trying to accomplish.

Now, if the report included some kind of experimentation, then the next part would be the experimental section. And this is a very important section because this is where you should put all the details that are necessary for anyone to reproduce your experiments based on the information in your report. But at the same time, it's a separate section because very often people who don't know yet whether they would be interested in what you were doing or not would just skip this section and go immediately to the results and discussion section.

That's the next section that we expect in a report, where you describe your findings, you interpret those findings, you discuss them, and explain why they're important, what is the significance of what you have done. Now at this stage, if the reader is, becomes interested in what you've done, they will go back, look at the experimental part, study the details of the experiment and then they will have a better understanding. If they decide that this is not something that they are really interested in, then they might just skip to the conclusions part. The conclusions is the next section.

Conclusions are not the same thing as summary, which comes at the beginning of the report. Conclusion should be as the name implies, what you've concluded

from your research. So think about what you're writing there, it cannot be just a summary of your report. It should be the most crucial and valiant points that stem from your findings.

And finally there is a recommendation section. This is sometimes tricky. We do expect that based on what you've done, you should be able to recommend some future steps or directions. That is not always easy to do but this is where you can ask your supervisor at the work place to help you with those future recommendations.

This is of course, followed by the list of literature which has to be first of all, uniform. One of the most common errors is that every reference is a different format. Students use, say, abbreviated journal names in one reference and then immediately below they use a non-abbreviated name. Sometimes they will put year first, then volume, then they flip it around. So, it's important, again, that you don't just copy and paste the references, that you actually go through them carefully and make sure that they have a uniform and consistent format. Again, if you submit a paper to a journal, if the format of the references is incorrect, the editor will send you the paper back immediately because you must be able to follow the guidelines, and that's an exercise in doing this.

Professionalism is something that is based on impression, really. Sometimes all it takes is a single look at the report and you can tell that whoever wrote it didn't really put his best foot forward. You open the report, and you see margins that are like five centimetres on each side, and you see double-and-a-half spacing and the report is maybe ten pages long, well obviously this is not professional. This is not what you expect from a serious effort.

I'm not saying that a report has to be 30 pages long, that's not the point. The point is that if you have something to tell, you have to be able to tell it in a way that conveys the message and at the same time, conveys respect for the reader. So content is important, but the form is also important.

This is not just our whimsy that we want you to follow the guidelines to the letter. This is simply making sure that you pay attention and that you can do this if required and that you can do a good job sticking to the guidelines and

conveying your message – at the same time, creating a great impression.

This is the first thing that people will be looking at. They flip through the report very often, and I'm not talking about just co-op reports, I'm talking about any kinds of reports, consultant report for example. They will just flip through it quickly and based on that flip that lasts maybe a few seconds, they will create a first impression. And that will immediately steer the attitude of the reader towards what they are reading. If they have the impression that this report is really not a piece of good work, then they will read the report with this kind of attitude, unfortunately. And that's unfortunate for the writer of the report.

So making sure that the report creates a good first impression is extremely important. And then making sure that you do your job, proofreading it, language-editing it, because this is how you show respect to whoever the audience for your report is. They want to be convinced that you tried to do your best job when preparing the work for them.

[Text only; no audio] Following all of the smaller guidelines and rules for report writing also shows your employer and your evaluator that you are professional.

[Barbara Moffatt]: Hello, my name is Barb Moffatt. I'm a professor in the Department of Biology and I've read a lot of reports for science over the years. And I have a few tips to help you do your best. The first one is, read the guidelines. Now, that's not always exciting, but this is really going to help you put together a strong report. In addition to that, I've got a few tips of the points that students tend to overlook from those guidelines.

The first one is to include a legend for all your figures and tables. Just as you see in a magazine or online associated with images or data, you'll see a short caption telling me what the picture is all about or the data. The same is true in your report, but these will probably be a little longer because you need to include the number of replicates, the scale bars, the treatments that were used. So the reader should be able to understand what is being transmitted in the picture or the table without consulting the text. That should be your bottom line when you're checking over your legend.

Another thing to keep in mind, is to remember to cite all copyrighted material that you take from other sources. This could be from a pamphlet, could be from a website, could be from a book or a publication, it doesn't matter. But you have to give attribution to the person who created it. And, in addition to that in-text citation, you have a full citation at the end in your reference list. Both are critical.

Another thing to keep in mind, is it always seems easy and convenient to just take your figures from another review article or publication without really putting a lot of your own thought into it. This is a big mistake. Any figures you use that are created by someone else really should be synthesized by you and incorporated into your argument and into your research. So you want to think carefully about when you decide to use that figure and whether they're really necessary or central to your story. Synthesize what they're saying, and possibly even recreate it so it matches your story and your information.

A few thoughts about paragraphs. Number one, what is a paragraph? It's to tell you information about one specific topic. So, if you look through your writing and find that your paragraphs extend for pages and pages, rethink where the ideas break and where you can contain those in specific paragraphs. Much easier as a reader. Think of yourself when you get a huge email, no paragraph breaks, it's hard to even take a big breath to get started.

So, think of your topic sentence – often very good to put that first so that the reader knows what's coming up – and then, sentences that are to the point, concise, and always relevant to the topic in your paragraph, and then close it out, leading to the next topic with some type of transition. That makes it a lot easier for your reader to understand your message and to stay interested as they read.

Finally, one little point that may be a specific to science, or little more common in science, but this is starting the sentence with a numeral. Do not do this! It's very easy to either restructure your sentence so you don't want the number to be at the very first item in the sentence, or write it out in full words. But do not start sentences with a numeral.

As illustrated by the faculty members, the structure of your technical report is extremely important because when formatted properly, it helps guide the reader through the content in a predictable and clear manner.

Sample Technical Reports

Examples of technical reports written by students employed during the co-op term:

- **Myles Reichert (Arts) - Website Data Analysis and Maintenance Needs (PDF)**
- **Richard (Zhi Fei) Dong (Life Physics) - Situation Report Regarding the 2019 Novel Coronavirus Outbreak (PDF)**
- **Lodi Sinanios (Biology) - The Importance of Diligence When Handling and Receiving Patient Information and Documentation (PDF)**
- **Ayon Bakshi (Computer Science) - Using Bounding Volume Hierarchies for Time and Resource Efficient Calculations (PDF)**
- **Yiqiao Sun (Mathematics) - Dynamic Data Storing Solution in Data Management System (PDF)**
- **Tracyl Ah-Leung (Chemical Engineering) - Coconut Project - Harvest Premix Line (PDF)**
- **Jessica R (Nanotechnology Engineering)- A Comparative Analysis of Mobile Software Development Kit Options for Indoor Mapping (PDF)**

Example of technical reports of students unemployed during co-op:

- **Enoch Fok (Mathematics/FARM) - Enhancing Training of Employees in the Workplace (PDF)**
- **Garen Simpson (Materials and Nanoscience) - The Applications and Importance of Glycogen-Based**

Nanoparticles in in Cancer Therapy (PDF)

- **Neelanjay Anthwal (Biology) - Effects of Different Microbiotic Growth in the Symbiotic Culture of Bacteria and Yeast on Kombucha Storage, Flavouring and Bottling (PDF)**

DISCLAIMER: These example reports are excellent samples of well-written technical reports but are not perfect. Please be sure to verify current referencing, citation, and formatting requirements for your own reports.

Section 1 Conclusion

In this section you learned the detailed structure of a technical report, the PD 11 guidelines, and faculty member perspectives on the importance of proper technical report structuring. As you begin to write your technical report use the **First Draft of Technical Report Template (DOCX)**. You are encouraged to revisit this section to ensure you are following the **PD 11 Technical Report Guidelines (PDF)** for how to structure your technical report.